

Microscopic Thin-Film and Biofilm Models for Bioregenerative Life Support

Verified MELiSSA Baselines, Thin-Film Algal Biofilm Algorithms, and a Control-Oriented Mathematical Synthesis

Compiled technical report

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Abstract

The best present answer to the question “what is the strongest thin-film microbial breakthrough for astronautical bioregenerative life support?” is not a single device. It is a verified modeling stack. The most astronautically verified microbial models are the MELiSSA photobioreactor and nitrifying fixed-bed biofilm lineages: they have mechanistic mass balances, light-transfer or biofilm-diffusion submodels, calibrated kinetics, pilot-plant validation, and, for the *Limnospira indica* photobioreactor model, interpretation of an International Space Station membrane-photobioreactor experiment. The strictest thin-film algal-biofilm mathematics is more often found in terrestrial attached-growth systems and open partial-differential-equation simulations; these are algorithmically transparent and relevant to reactor design, but their space validation is weaker. This report therefore treats MELiSSA as the validated systems-engineering baseline and treats thin-film algal-biofilm models as the most promising mathematical design layer to be coupled into that baseline.

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1 Scope, evidence standard, and main conclusion

This report synthesizes seven candidate breakthrough areas for microscopic, thin-film, and biofilm life-support research. The term BRLSS is used broadly for bioregenerative life-support systems; BLSS is used when the emphasis is specifically on regenerative spacecraft or habitat support. The key distinction is between two kinds of evidence:

1. **Astronautical verification:** model predictions and control logic have been tested in a life-support context, at MELiSSA pilot scale, under integrated animal-compartment operation, or in a flight experiment.
2. **Thin-film mathematical transparency:** the attached microbial layer is represented explicitly by reaction–diffusion, mixture theory, active-layer depth, light attenuation, harvesting, or biofilm-thickness dynamics.

The strongest single starting point is the MELiSSA C4a *Limnospira indica* photobioreactor model, because it combines radiative transfer, light-transfer-induced physiology, carbonate chemistry, hydrodynamic assumptions, gas-liquid mass transfer, pilot-scale operation, and ISS flight-experiment interpretation (Poughon et al. 2021, 1–3, 6, 8–9). The most directly “biofilm” MELiSSA component is C3, the nitrifying packed/fixed-bed biofilm reactor containing attached *Nitrosomonas europaea* and *Nitrobacter winogradskyi*; its reactor model was developed for steady and dynamic operation, and later integration work showed robust ammonium conversion under coupled operation (Pérez et al. 2005, 2359–2369)(Garcia-Gragera et al. 2021, 2, 11, 13). The most inspectable strict thin-film algal-biofilm mathematics appears in terrestrial rotating or scraped phototrophic biofilm systems and open 2D mixture-theory simulations (Polizzi et al. 2022, 1, 3–5)(Blanken et al. 2014, 2436–2445).

The conclusion is therefore deliberately nuanced. Microbial biofilm and thin-film models are mature enough for rigorous reactor-level design and control. However, the fully space-verified examples remain mostly MELiSSA-style fixed-bed nitrification and suspended or membrane photobioreactor cultivation, not fully space-tested algal thin-film panels. The most defensible research direction is to combine the open thin-film biofilm PDE models with MELiSSA-style mass-balanced control architecture.

2 Nomenclature guide

Table 1: Symbols used in the report. Units vary by implementation; default concentration units are mass or mole per reactor volume unless stated otherwise.

Symbol	Meaning	Typical units
A, N, E, L	Active algal cells, non-growing algal cells, extracellular polymeric substances, and liquid volume fractions in a photosynthetic biofilm mixture model	dimensionless volume fraction
C, O, S	Dissolved inorganic carbon, oxygen, and nitrate fields in the photosynthetic biofilm model	mol m^{-3}

Symbol	Meaning	Typical units
C_i	Concentration of soluble species i in bulk liquid or biofilm pore liquid	mol m^{-3} or kg m^{-3}
$C_{\text{NH}_4}, C_{\text{NO}_2}, C_{\text{NO}_3}, C_{\text{O}_2}$	Ammonium, nitrite, nitrate, and dissolved oxygen concentrations in the nitrifying biofilm reactor	usually N or O_2 concentration
C_T or TIC	Total inorganic carbon, $[\text{CO}_2] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$	mol m^{-3}
c	Radiative-transfer geometric correction, commonly $c = \min(1, L_0/L)$ in the <i>Limnospira indica</i> two-flux model	dimensionless
D	Dilution rate or, with subscript i , molecular diffusion coefficient	d^{-1} or $\text{m}^2 \text{s}^{-1}$
E_a, E_s, b	Mass absorption coefficient, mass scattering coefficient, and scattering backward fraction in the two-flux photobioreactor model	$\text{m}^2 \text{kg}^{-1}$, $\text{m}^2 \text{kg}^{-1}$, dimensionless
$F_{\text{in}}, F_{\text{out}}$	Inlet and outlet gas flow rates	L h^{-1} or mol s^{-1}
G_z, G_r	Local hemispherical irradiance in flat and cylindrical geometries	$\mu\text{mol m}^{-2} \text{s}^{-1}$
$I(z)$	Light intensity at depth z in a biofilm or culture	$\mu\text{mol m}^{-2} \text{s}^{-1}$
$J_{L,i}, J_{G,i}$	Liquid diffusive flux and gas-liquid transfer flux of species i	concentration flux
K_i	Half-saturation or inhibition constant for species i	concentration
k_{La}	Volumetric liquid-side mass-transfer coefficient	s^{-1} or h^{-1}
L	Optical path length or biofilm height, depending on context	m
L_F	Biofilm thickness in the nitrifying fixed-bed reactor model	m
MW_{O_2}	Molecular weight of oxygen	g mol^{-1}
$P/2e^-$	Photon requirement per two electrons transferred in the <i>Limnospira indica</i> light physiology model	dimensionless
q_0	Incident light flux density at the photobioreactor boundary	$\mu\text{mol m}^{-2} \text{s}^{-1}$
q_C, q_N, q_{O_2}	Specific consumption or production rates for carbon, nitrogen, or oxygen	concentration per biomass per time
Q	Liquid volumetric flow rate	$\text{m}^3 \text{s}^{-1}$ or L d^{-1}
r_j	Volumetric rate of biological process j	$\text{mol m}^{-3} \text{s}^{-1}$
r_{O_2}	Volumetric oxygen production rate in the integrated pilot photobioreactor	$\text{gO}_2 \text{L}^{-1} \text{h}^{-1}$ or equivalent

Symbol	Meaning	Typical units
u_F, u_{det}	Solids growth velocity and detachment velocity in a biofilm	m s^{-1}
V_R	Reactor liquid volume	L or m^3
$X, X_{\text{Nts}}, X_{\text{Ntb}}$	Biomass concentration; biomass of <i>Nitrosomonas europaea</i> and <i>Nitrobacter winogradskyi</i>	kg m^{-3}
Y_j	Yield coefficient for biological process j	biomass/substrate
α, δ, γ	Two-flux radiative-transfer parameters combining absorption, scattering, geometry, and biomass concentration	variable
η_j	Closure, recovery, or utilization efficiency for species or service j	dimensionless
μ, μ_{max}	Specific growth rate and maximum specific growth rate	d^{-1}
ν_{ij}	Stoichiometric coefficient of species i in process j	dimensionless
N	Stoichiometric matrix mapping process rates into species balances	dimensionless

3 Ranked landscape of breakthroughs

Table 2: Best-matching model breakthroughs for astronautical microbial thin-film and biofilm BLSS research.

Rank	Breakthrough area	Why it matters for astronautical BLSS	Accessible mathematical content	Verification assessment
1	MELiSSA C4a <i>Limnospira indica</i> photobioreactor model	Best single starting point for a verified microbial oxygen/biomass compartment. It covers light delivery, growth, oxygen production, carbon chemistry, and scale transfer from pilot plant to ISS microreactor interpretation.	Two-flux radiative transfer, light-transfer-induced physiology, stoichiometric growth, carbonate equilibrium, hydrodynamic tank-in-series assumptions, gas-liquid balances, operating maps for control.	Strong. The model was applied to an 80 L pilot reactor and a 50 mL ISS membrane photobioreactor experiment (Poughon et al. 2021, 1, 6, 8–9).
2	MELiSSA C3 nitrifying fixed-bed biofilm reactor	Most directly attached-biofilm component of the MELiSSA loop. It converts ammonium waste to nitrate nutrient while avoiding nitrite accumulation.	Radial biofilm diffusion, multispecies kinetics, oxygen transfer, biofilm growth/detachment, tanks-in-series hydrodynamics, calibrated nitrifier kinetics.	Strong ground verification. The fixed-bed model was calibrated/validated for steady and dynamic modes, and integration studies reported robust ammonium conversion (Pérez et al. 2005, 2359–2369)(Montràs i Boet 2009, 96–115)(Garcia-Gragera et al. 2021, 11, 13).
3	Integrated MELiSSA pilot plant: C3 + C4a + animal crew mock-up	Shows models functioning as life-support system engineering, not isolated culture productivity. It links nitrogen closure, oxygen production, carbon dioxide uptake, and animal respiration.	Compartment mass balances, gas-phase oxygen/CO ₂ equations, light-intensity control, nitrogen balance, microbial load changes, disturbance recovery.	Very strong ground integration: continuous C3–C4a operation over 276 days and gas integration with animal compartment over 111 days (Garcia-Gragera et al. 2021, 5–6, 8, 11, 13).

Rank	Breakthrough area	Why it matters for astronomical BLSS	Accessible mathematical content	Verification assessment
4	MELiSSA system-level stoichiometric and dynamic loop models	Provides mission sizing and closure bookkeeping across crew, waste, gases, water, microbes, crops, and food.	Stoichiometric matrices, element balances, linear/nonlinear mass-balance constraints, compartment process maps.	Moderate-to-strong. Several submodels have pilot evidence; complete autonomous human-scale closure remains a target rather than a finished demonstration (Vermeulen et al. 2023, Abstract, sec. 5).
5	<i>Chlorella vulgaris</i> / PBR@LSR hybrid life-support and ESM analysis	Useful for NASA/ESA-style trade studies of oxygen, food substitution, power, cooling, volume, launch mass, and crew time.	Equivalent system mass, productivity-to-reactor-volume sizing, food-substitution fraction, oxygen-production sizing, break-even mission duration.	Strong but narrower. ISS algal cultivation and long-duration ground cultivation exist, but public first-principles reactor models are less exposed than the MELiSSA <i>Limnospira indica</i> line (Detrell 2021, 3–5, 7–8).
6	Rotating or attached algal biofilm photobioreactors, including Algadisk-style systems	Directly relevant to strict thin-film microbial life because biomass is attached, harvestable, and potentially compact with reduced water inventory.	Light-limited biofilm growth models, day/night operation, reactor staging, CO ₂ utilization, harvesting interval and biofilm-thickness effects.	Strong terrestrial verification, weak space verification. Rotating biofilm tests achieved high areal productivity and long retention, but are not yet flight-validated BLSS hardware (Blanken et al. 2014, 2436–2445)(Blanken 2016, abstract).

Rank	Breakthrough area	Why it matters for astronomical BLSS	Accessible mathematical content	Verification assessment
7	Open 2D PDE / mixture-theory photosynthetic biofilm simulation	Clearest algorithmic route for strict thin-film design questions: light penetration, nutrient gradients, active-layer depth, harvesting interval, and patterning.	Multi-component mass and momentum balances, reaction–diffusion–advection equations, light and substrate limitation, mortality, extra-cellular matrix, gas/liquid boundary conditions.	Good mathematical verification, limited BLSS verification. Treat as a rigorous design model to be adapted into validated MELiSSA-style architecture (Polizzi et al. 2022, 1, 3–5).

4 Common mathematical backbone for microbial BLSS compartments

Even when the hardware differs, the models share a small set of mathematical structures. A compartment-level state vector

$$\mathbf{x}(t) = \begin{bmatrix} m_{\text{O}_2} & m_{\text{CO}_2} & m_{\text{NH}_4} & m_{\text{NO}_2} & m_{\text{NO}_3} & m_X & \cdots \end{bmatrix}^T \quad (1)$$

can be written as

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}_{\text{in}}(t) - \mathbf{F}_{\text{out}}(t) + V\mathbf{N}^T \mathbf{r}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) + \mathbf{T}_g(\mathbf{x}, p, T) + \mathbf{d}(t), \quad (2)$$

where $\mathbf{N}$ is a stoichiometric matrix, $\mathbf{r}$ contains biological rates, $\mathbf{u}$ contains manipulated variables such as light flux, dilution rate, gas flow, or feed concentration, $\boldsymbol{\theta}$ contains kinetic and transport parameters, and $\mathbf{T}_g$ accounts for gas-liquid exchange. Equation (2) is not a single published equation; it is a compact synthesis of the mass-balanced compartment logic used in the MELiSSA photobioreactor, nitrifying biofilm, and integration literature (Poughon et al. 2021, 2–3, 6)(Garcia-Gragera et al. 2021, 5–6)(Montràs i Boet 2009, 96–115).

For system closure, a static stoichiometric model can be written as

$$\mathbf{A}\boldsymbol{\xi} = \mathbf{0}, \quad (3)$$

$$\mathbf{B}\boldsymbol{\xi} + \mathbf{s}_{\text{crew}} + \mathbf{s}_{\text{stores}} = \mathbf{0}, \quad (4)$$

where $\mathbf{A}$ maps reactions to conserved elements such as C, H, O, and N, $\boldsymbol{\xi}$ is a vector of reaction extents or process throughputs, and $\mathbf{B}$ maps process throughputs to life-support commodities. A dynamic control layer then enforces acceptable bounds:

$$\mathbf{x}_{\min} \leq \mathbf{x}(t) \leq \mathbf{x}_{\max}, \quad \mathbf{u}_{\min} \leq \mathbf{u}(t) \leq \mathbf{u}_{\max}. \quad (5)$$

The 2023 MELiSSA stoichiometric review is important because it explicitly treats the loop as a mass-balanced, multi-compartment chemical and biological network and identifies which published models expose equations and stoichiometric coefficients (Vermeulen et al. 2023, Abstract, sec. 2–5). The limitation is equally important: static closure can be written at 100 percent in a model, but actual biological systems have leakage, recalcitrant products, digestibility limits, maintenance dynamics, and control failures (Vermeulen et al. 2023, sec. 5).

5 MELiSSA C4a *Limnospira indica* photobioreactor model

5.1 Why this is the strongest verified microbial modeling baseline

The C4a *Limnospira indica* compartment is the best single starting point because the model connects photobiology to astronautical operating variables. Poughon and colleagues describe the model as a mechanistic, mass-balanced approach combining radiative transfer and light-transfer-induced physiology, then apply it to an 80 L MELiSSA pilot photobioreactor and to a 50 mL ISS membrane photobioreactor experiment (Poughon et al. 2021, 1, 6, 8–9). The reactor function is not merely biomass production. In MELiSSA it is an air-revitalization and edible-biomass compartment: it consumes carbon dioxide, produces oxygen, and contributes microbial food or feedstock.

5.2 Radiative-transfer submodel

The core input to cyanobacterial physiology is not only incident light but the local light actually seen by cells. For a flat, one- or two-sided illuminated culture, the two-flux expression can be represented as

$$\frac{G_z}{q_0} = 2 \frac{n+2}{n+1} \frac{(1+\alpha) \exp[\delta(L-z)] - (1-\alpha) \exp[-\delta(L-z)]}{(1+\alpha)^2 \exp(\delta L) - (1-\alpha)^2 \exp(-\delta L)}, \quad (6)$$

while for a cylindrical reactor the analogous expression uses modified Bessel functions:

$$\frac{G_r}{q_0} = 2 \frac{n+2}{n+1} \frac{I_0(\delta r)}{I_0(\delta L) + \alpha I_1(\delta L)}. \quad (7)$$

Here q_0 is incident light flux density, G is local irradiance, L is optical path length, n is a phase-function exponent, and α and δ bundle absorption, scattering, and biomass effects. The published MELiSSA implementation uses mass absorption and scattering coefficients for *Limnospira indica* and a geometric correction

$$c = \min\left(1, \frac{L_0}{L}\right), \quad (8)$$

which prevents overinterpreting optical behavior outside the calibrated thickness regime (Poughon et al. 2021, 2). The biological implication is direct: any thin-film or panel model has to treat light as a spatial field, not as a scalar imposed on the whole reactor.

5.3 Light-transfer-induced physiology and growth

The biological model links the absorbed photon field to growth, storage, oxygen production, and biomass composition. A compact reactor-scale representation is

$$\frac{dX}{dt} = [\mu(\bar{G}, C_T, N, pH, T) - D - k_d] X, \quad (9)$$

$$\frac{dC_T}{dt} = D(C_{T,\text{in}} - C_T) - q_C(\bar{G}, C_T)X + k_L a_{\text{CO}_2}(C_{\text{CO}_2}^* - C_{\text{CO}_2}), \quad (10)$$

$$\frac{dC_{\text{O}_2}}{dt} = q_{\text{O}_2}(\bar{G}, C_T)X - DC_{\text{O}_2} + k_L a_{\text{O}_2}(C_{\text{O}_2}^* - C_{\text{O}_2}), \quad (11)$$

where D is the dilution rate, $\bar{G}$ is a culture-averaged light term derived from equations (6) and (7), C_T is total inorganic carbon, and q_C and q_{O_2} are specific carbon uptake and oxygen production functions.

The carbonate speciation layer is necessary because pH controls carbon availability:

$$C_T = [\text{CO}_2] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}], \quad (12)$$

$$[\text{CO}_2] = \alpha_0(pH)C_T, \quad [\text{HCO}_3^-] = \alpha_1(pH)C_T, \quad [\text{CO}_3^{2-}] = \alpha_2(pH)C_T. \quad (13)$$

The ISS application is especially instructive because pH was not directly measured or controlled in the microreactor; it had to be inferred from the model and gas-pressure behavior (Poughon et al. 2021, 8–9). This is a design warning for any future thin-film BLSS panel: a phototrophic model with elegant light physics but weak pH and carbonate accounting is not mission ready.

The MELiSSA model also uses a feasibility check based on the photon requirement per two electrons transferred, $P/2e^-$. One published relation links exopolysaccharide fraction to the light physiology variable:

$$x_{\text{EPS}} = 1.33 \left(\frac{P}{2e^-} - 1.23 \right). \quad (14)$$

Poughon and colleagues use this kind of constraint to separate mathematically possible model solutions from biologically feasible ones; this is important because control maps may otherwise include operating points where the equations solve but the cells cannot maintain the predicted metabolism (Poughon et al. 2021, 2, 4).

5.4 Hydrodynamics, gas transfer, and control maps

At pilot scale the reactor is not simply a well-mixed flask. The MELiSSA model represents the airlift photobioreactor using riser/downcomer assumptions, gas plug-flow treatment, carbonate equilibrium, and gas-liquid transfer terms (Poughon et al. 2021, 6). In reduced form, for a tank-in-series discretization $n = 1, \dots, N$,

$$\frac{dC_{i,n}}{dt} = \frac{Q}{V_n}(C_{i,n-1} - C_{i,n}) + \sum_j \nu_{ij}r_{j,n} + k_L a_i(C_{i,n}^* - C_{i,n}). \quad (15)$$

The same model family can be reduced to operating maps, sometimes called “abaci,” for control. For example, a controller can use light input q_0 , dilution rate D , and inlet gas composition to keep carbon limitation and oxygen output within an acceptable region:

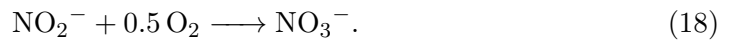
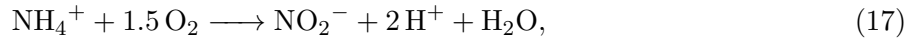
$$\mathcal{F}_{\text{feasible}} = \{(q_0, D, F_g, C_{T,\text{in}}) : C_T > C_{T,\text{min}}, pH < pH_{\text{max}}, P/2e^- < \Pi_{\text{max}}, r_{O_2} \geq r_{O_2,\text{req}}\}. \quad (16)$$

This is exactly the move that makes the model astronautical rather than merely biological: the output is a controllable operating envelope, not just a growth curve (Poughon et al. 2021, 7–9).

6 MELiSSA C3 nitrifying fixed-bed biofilm model

6.1 Why C3 is the most directly attached-biofilm breakthrough

The MELiSSA C3 reactor is a compact packed-bed/fixed-bed biofilm system. It hosts immobilized *Nitrosomonas europaea* and *Nitrobacter winogradskyi* that convert ammonium to nitrate. The biochemical purpose is nitrogen closure: ammonium-rich waste streams must be oxidized to nitrate without accumulating nitrite, because nitrate is the preferred nutrient form for downstream phototrophic or higher-plant compartments. Garcia-Gragera and colleagues describe the integrated C3 reactor as a 7 L cylindrical packed-bed with polystyrene beads and co-cultured nitrifiers forming a biofilm (Garcia-Gragera et al. 2021, 2). The oxygen demand for complete nitrification is approximately two moles of oxygen per mole of ammonium nitrogen oxidized (Garcia-Gragera et al. 2021, 2):



The 2005 fixed-bed model is a major breakthrough because it was written for both steady-state and dynamic operation and explicitly for a BLSS nitrogen compartment (Pérez et al. 2005, 2359–2369). The later UAB thesis exposes the model anatomy in greater detail: soluble species, particulate species, diffusion, biofilm growth, detachment, oxygen gas-liquid transfer, reactor hydrodynamics, and AQUASIM implementation (Montràs i Boet 2009, 96–115).

6.2 Radial diffusion and biofilm balance equations

Inside an attached spherical or bead-supported biofilm, soluble species obey Fickian diffusion plus reaction. In spherical coordinates a generic soluble species balance is

$$\frac{\partial C_i}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_i \frac{\partial C_i}{\partial r} \right) + \sum_j \nu_{ij} r_j(C_i, X_k), \quad (19)$$

with liquid diffusive flux

$$J_{L,i} = -D_i \frac{\partial C_i}{\partial r}. \quad (20)$$

Montràs reports this Fick-law treatment, biofilm matrix/pore-water partition, and radial biofilm discretization as part of the packed-bed model (Montràs i Boet 2009, 96–97, 114–115). Biofilm solids move because cells grow, decay, detach, and displace one another. A compact representation is

$$\frac{\partial X_k}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 u_F X_k \right) = R_{X,k}, \quad (21)$$

$$\frac{dL_F}{dt} = u_F - u_{\text{det}}, \quad (22)$$

where L_F is film thickness, u_F is net growth velocity, and u_{det} is detachment velocity. This is the core difference between an attached-growth BLSS reactor and a suspended culture: the local reaction environment is a spatially structured film.

Oxygen transfer enters through the bulk liquid as

$$J_{G,O_2} = k_L a_{O_2} (C_{O_2}^* - C_{O_2}), \quad (23)$$

which must be coupled to oxygen consumption by nitrifiers (Montràs i Boet 2009, 97–98). Operationally, this is why C3 control cannot be reduced to ammonium feed alone; dissolved oxygen steps can create nitrite accumulation if *Nitrobacter winogradskyi* becomes oxygen-limited.

6.3 Nitrifier kinetics

A compact version of the calibrated nitrifier kinetics is

$$r_{\text{Nts}} = \mu_{\text{Nts}} X_{\text{Nts}} \frac{C_{\text{NH}_4}}{K_{\text{NH}_4} + C_{\text{NH}_4} + C_{\text{NH}_4}^2 / K_{I,FA,\text{Nts}}} \frac{C_{\text{O}_2}}{K_{\text{O}_2,\text{Nts}} + C_{\text{O}_2}}, \quad (24)$$

$$r_{\text{Ntb}} = \mu_{\text{Ntb}} X_{\text{Ntb}} \frac{C_{\text{NO}_2}}{K_{\text{NO}_2} + C_{\text{NO}_2} + C_{\text{NO}_2}^2 / K_{I,FNA,\text{Ntb}}} \frac{C_{\text{O}_2}}{K_{\text{O}_2,\text{Ntb}} + C_{\text{O}_2}} I_{FA,\text{Ntb}}. \quad (25)$$

At 28 °C and pH 8, the thesis reports representative kinetic values such as $K_{\text{NH}_4} = 0.0175$, $K_{\text{NO}_2} = 5.04 \times 10^{-3}$, $K_{\text{O}_2,\text{Nts}} = 1.616 \times 10^{-4}$, $K_{\text{O}_2,\text{Ntb}} = 5.44 \times 10^{-4}$, $K_{I,FA,\text{Nts}} = 0.116$, $K_{I,FNA,\text{Ntb}} = 1.56 \times 10^{-4}$, $\mu_{\text{Nts}} = 1.368 \text{ d}^{-1}$, and $\mu_{\text{Ntb}} = 0.864 \text{ d}^{-1}$ (Montràs i Boet 2009, 112). The exact units depend on the model concentration basis, but the structure is clear: ammonium oxidation and nitrite oxidation are limited by substrate and oxygen and inhibited by free ammonia or free nitrous acid.

6.4 Reactor hydrodynamics and numerical implementation

The fixed-bed reactor can be represented as a series of bulk-liquid compartments coupled to local biofilm modules:

$$\frac{dC_{i,n}^{\text{bulk}}}{dt} = \frac{Q}{V_n} (C_{i,n-1}^{\text{bulk}} - C_{i,n}^{\text{bulk}}) - a_n J_{L,i}|_{r=L_F} + k_L a_i (C_i^* - C_{i,n}^{\text{bulk}}), \quad (26)$$

where a_n is a support-area-to-volume factor for compartment n . Montràs describes a seven-compartment tank-in-series representation estimated from residence-time-distribution data and implemented in AQUASIM with bulk liquid, biofilm matrix, and pore-water compartments (Montràs i Boet 2009, 105, 114–115). This gives enough information to build a reproducible method-of-lines model even without the original AQUASIM files.

7 Integrated MELiSSA pilot plant: from reactor equations to life-support control

7.1 Integration is the astronautical systems breakthrough

The integrated MELiSSA pilot plant matters because it couples microbial subsystems under control. Garcia-Gragera and colleagues connected the nitrifying compartment, the photosynthetic compartment, and an animal compartment in staged integration. The liquid C3–C4a connection ran in continuous mode for 276 days, and gas integration between the photobioreactor and the animal compartment ran for 111 days (Garcia-Gragera et al. 2021, 5, 8). During the integrated campaign, ammonium conversion was maintained at 100 percent for roughly 200 days despite load changes and disturbance events, with nitrite peaks recovering after operational adjustments (Garcia-Gragera et al. 2021, 11, 13).

This is the mathematical threshold that thin-film BLSS prototypes must meet. A thin biofilm panel is not astronautically convincing merely because it grows biomass per area. It must accept coupled disturbances: crew CO_2 production, O_2 demand, variable nitrogen loading, pH drift, light limits, gas exchange, and microbial stability.

7.2 Oxygen-production balance used in integration

The integrated photobioreactor oxygen production can be computed from inlet and outlet gas flows. A representative disconnected-gas expression is

$$r_{\text{O}_2} = \frac{(F_{\text{in}} - F_{\text{CO}_2})0.2089 - F_{\text{out}}y_{\text{O}_2,\text{out}}}{22.4 MW_{\text{O}_2} V_R}, \quad (27)$$

where 0.2089 is the molar fraction of oxygen in air, F_{CO_2} accounts for carbon dioxide feed contribution, $y_{\text{O}_2,\text{out}}$ is oxygen molar fraction in the outlet, and V_R is reactor volume. In connected operation the measured inlet oxygen fraction is used:

$$r_{\text{O}_2} = \frac{F_{\text{in}}y_{\text{O}_2,\text{in}} - F_{\text{out}}y_{\text{O}_2,\text{out}}}{22.4 MW_{\text{O}_2} V_R}. \quad (28)$$

These equations are a useful bridge from biology to crew support because they convert a photobioreactor state into an animal-compartment oxygen service (Garcia-Gragera et al. 2021, 5).

7.3 A reduced control structure

A simple control abstraction for the C4a–animal gas loop is

$$e_{\text{O}_2}(t) = y_{\text{O}_2,\text{sp}} - y_{\text{O}_2,\text{animal}}(t), \quad (29)$$

$$q_0(t) = \text{sat}_{[q_{\min}, q_{\max}]} \left(q_{0,0} + K_P e_{\text{O}_2}(t) + K_I \int_0^t e_{\text{O}_2}(\tau) d\tau \right), \quad (30)$$

$$D(t) = \text{sat}_{[D_{\min}, D_{\max}]} (D_0 + K_N (C_{\text{NO}_3,\text{sp}} - C_{\text{NO}_3})). \quad (31)$$

The exact MELiSSA control architecture is richer, but equations (29) and (30) captures the principle: light, feed, and gas flow are manipulated to make a biological compartment deliver a life-support service. Garcia-Gragera and colleagues report that the photobioreactor oxygen production was able to fulfill the rat-compartment oxygen need during the gas-integration period, while the nitrifying reactor maintained ammonium conversion and recovered from nitrite events (Garcia-Gragera et al. 2021, 11, 13).

8 System-level stoichiometry and equivalent system mass

8.1 Stoichiometric closure models

At mission scale, the question is not only “does the film grow?” It is “does the loop close with acceptable mass, energy, volume, time, and reliability?” A stoichiometric closure model treats all compartments as a linked reaction network. A simplified element balance for C, H, O, and N is

$$\begin{bmatrix} \text{C} & \text{H} & \text{O} & \text{N} \end{bmatrix}_{\text{in}} + \mathbf{A}\boldsymbol{\xi} = \begin{bmatrix} \text{C} & \text{H} & \text{O} & \text{N} \end{bmatrix}_{\text{out}}. \quad (32)$$

A closure metric for commodity j can be written as

$$\eta_j = 1 - \frac{F_{j,\text{resupply}} + F_{j,\text{loss}}}{F_{j,\text{demand}}}, \quad (33)$$

where $\eta_j = 1$ is ideal closure and $\eta_j = 0$ means no regenerative contribution. The MELiSSA stoichiometric review is valuable because it surveys how published compartment equations and stoichiometric coefficients can be assembled into a whole-loop mass model (Vermeulen et al. 2023, Abstract, sec. 2–5).

8.2 Equivalent system mass for algal photobioreactors

NASA- and ESA-style mission trades often use equivalent system mass:

$$\text{ESM} = M + \lambda_V V + \lambda_P P + \lambda_Q Q + \lambda_t t_{\text{crew}}, \quad (34)$$

where M is mass, V is volume, P is power, Q is cooling or heat-rejection load, t_{crew} is crew time, and λ terms convert non-mass penalties to equivalent mass. A bioregenerative subsystem is favorable when

$$\text{ESM}_{\text{biological}}(T) < \text{ESM}_{\text{physicochemical}}(T) \quad (35)$$

for mission duration T .

Detrell’s *Chlorella vulgaris* lunar-base analysis is important because it makes this trade explicit. The article estimates that 50 L to 100 L of reactor volume per person could provide roughly 10–30 percent food substitution, and break-even durations were reported in the approximate range of 4.0–6.5 years for a 30 percent food case and 4.8–7.3 years for a 10 percent food case (Detrell 2021, 3–5). The same review cautions that most space photobioreactor experiments are short and small, and that biofilm formation and long-term stability must be understood before full life-support use (Detrell 2021, 7–8). This is an important warning for thin-film systems: attachment can be a design feature when intentional, but an operational fault when it clogs or fouls a suspended-culture photobioreactor.

9 Strict thin-film algal biofilm models

9.1 Rotating and attached phototrophic biofilms

Attached microalgal films are attractive for BLSS because they can reduce liquid inventory, simplify harvesting, increase areal productivity, and separate biomass retention from hydraulic residence time. In rotating biological contactor or AlgaDisk-style designs, a thin algal film alternates between liquid nutrient exposure and gas/light exposure. Blanken and colleagues demonstrated *Chlorella sorokiniana* growth in a rotating biological contactor photobioreactor with sustained attached cultivation, while the thesis line develops models for biofilm growth, day/night operation, CO₂ utilization, and staged operation (Blanken et al. 2014, 2436–2445)(Blanken 2016, abstract).

A minimal areal biofilm production model can be written as

$$\frac{\partial X(z, t)}{\partial t} = \mu(I(z, t), C(z, t), N(z, t))X - k_d X, \quad (36)$$

$$I(z, t) = I_0(t) \exp \left[- \int_0^z \kappa_X X(\zeta, t) + \kappa_E E(\zeta, t) d\zeta \right], \quad (37)$$

$$P_A(t) = \int_0^{h(t)} [\mu(I, C, N) - k_d] X(z, t) dz, \quad (38)$$

$$h(t_n^+) = h_H \quad \text{after harvest at } t = t_n. \quad (39)$$

Here P_A is areal productivity and h_H is post-harvest film thickness. This class of model is directly relevant to spacecraft design because the state variable of interest is not only cell concentration but film thickness and illuminated active depth.

9.2 Open 2D PDE and mixture-theory simulations

Polizzi and colleagues provide the clearest open algorithmic route for strict photosynthetic biofilm modeling. Their model represents active cells, non-growing cells, extracellular matrix, and liquid as volume fractions; dissolved carbon, oxygen, and nitrate diffuse and react; kinetics are regulated by light, nitrate, oxygen, and inorganic carbon (Polizzi et al. 2022, 1, 3–5). In compact notation,

$$\frac{\partial \phi_k}{\partial t} + \nabla \cdot (\phi_k \mathbf{v}_k) = R_k(\phi, C, O, S, I), \quad k \in \{A, N, E, L\}, \quad (40)$$

$$\sum_{k \in \{A, N, E, L\}} \phi_k = 1, \quad (41)$$

$$\frac{\partial \psi}{\partial t} + \nabla \cdot (\psi \mathbf{v}_L) - \nabla \cdot (D_\psi \nabla \psi) = R_\psi(\phi, C, O, S, I), \quad \psi \in \{C, O, S\}. \quad (42)$$

A useful light closure is

$$I(x, z, t) = I_0(x, t) \exp \left[- \int_0^z (\kappa_A \phi_A + \kappa_N \phi_N + \kappa_E \phi_E)(x, \zeta, t) d\zeta \right]. \quad (43)$$

Boundary conditions couple the film to gas or medium:

$$-D_C \nabla C \cdot \mathbf{n} = k_C(C - C^*), \quad (44)$$

$$-D_O \nabla O \cdot \mathbf{n} = k_O(O - O^*), \quad (45)$$

$$S = S_{\text{medium}} \quad \text{on wetted nutrient boundaries.} \quad (46)$$

The value of this model family for BLSS is that it exposes design variables hidden by well-mixed models: active-layer depth, substrate-gradient failure, oxygen inhibition, extracellular matrix

accumulation, and harvesting pattern. Polizzi and colleagues report that an active region of roughly 75 μm , optimized harvesting at roughly week scale, and patterned harvesting can materially affect productivity (Polizzi et al. 2022, 1, 5, 15–16). The astronautical limitation is that this is a terrestrial design model, not a flight-validated life-support compartment.

9.3 How the thin-film PDE layer should be coupled to MELiSSA control

The most promising synthesis is a two-level model:

1. a high-fidelity biofilm PDE that predicts areal oxygen production, carbon uptake, biomass harvest, nitrogen demand, and fouling risk as functions of light, wetting, gas composition, film thickness, and harvesting; and
2. a reduced-order MELiSSA-style compartment model that embeds those predictions into mass-balance control for crew oxygen, carbon dioxide, ammonium, nitrate, water, and food.

Mathematically, the PDE can be projected into a low-dimensional service model:

$$\begin{bmatrix} R_{\text{O}_2} \\ R_{\text{CO}_2} \\ R_{\text{NO}_3} \\ R_X \\ R_{\text{risk}} \end{bmatrix} = \mathcal{R}_{\text{ROM}}(q_0, F_g, Q, h, t_{\text{harvest}}, \theta_{\text{film}}), \quad (47)$$

where $\mathcal{R}_{\text{ROM}}$ is a reduced-order map trained or fitted from PDE simulations and experiments. The compartment-level controller then uses

$$\frac{d\mathbf{x}_{\text{habitat}}}{dt} = \mathbf{f}_{\text{crew}}(t) + \mathbf{B}\mathcal{R}_{\text{ROM}}(\mathbf{u}, t) - \mathbf{g}_{\text{losses}}(t). \quad (48)$$

This approach preserves the design value of thin-film spatial models while avoiding the computational burden of solving a full 2D or 3D biofilm PDE inside every mission-level controller.

10 Validation ladder for a future thin-film BLSS panel

A rigorous validation program should not jump directly from a productive terrestrial biofilm to a flight claim. A credible ladder is:

1. **Bench PDE calibration:** estimate light attenuation, diffusion, uptake, EPS, mortality, and harvesting parameters using controlled films.
2. **Closed-gas service tests:** show predictable oxygen production and carbon dioxide uptake from gas balances similar to equations (27) and (28).
3. **Nitrogen coupling:** link the phototrophic film to nitrified waste streams and measure nitrate uptake, pH drift, and nitrite sensitivity.
4. **Dynamic disturbance tests:** impose crew-like CO_2 loads, day/night cycles, flow interruptions, pH disturbances, and harvesting events.
5. **Reduced-gravity risk tests:** characterize wetting, bubble behavior, capillary retention, gas exchange, film sloughing, and sensor reliability.

6. **Integrated pilot closure:** connect the panel to a C3-like nitrifying unit and a crew metabolic simulator.
7. **Flight demonstration:** operate a small film reactor with measured gas exchange, pH/carbonate accounting, imaging of film morphology, and postflight biomass composition.

The ladder is intentionally patterned after the MELiSSA precedent: model, pilot verification, integration, and flight interpretation. The *Limnospira indica* model demonstrates this path for a suspended/membrane photobioreactor (Poughon et al. 2021, 1, 6, 8–9); the C3 reactor demonstrates it for attached nitrifying biofilms in ground integration (Garcia-Gragera et al. 2021, 11, 13); the thin-film algal literature supplies the missing spatial algorithms (Polizzi et al. 2022, 3–5)(Blanken et al. 2014, 2436–2445).

11 Research gaps and design recommendations

11.1 Gaps

1. **Public code gap.** The most verified MELiSSA models are described in papers and theses, but complete reusable code is not generally available. The Polizzi model is more open algorithmically, but less flight-validated (Polizzi et al. 2022, 5).
2. **Partial-gravity gap.** ISS microgravity and 1-g ground validation do not automatically resolve lunar or Martian partial-gravity wetting, bubble, and film-stability behavior.
3. **Biofilm control gap.** In C3, biofilm is the reactor. In suspended PBRs, unwanted biofilm formation can become fouling. A thin-film algal BLSS system must distinguish controlled attachment from failure-mode attachment.
4. **pH/carbonate gap.** The *Limnospira indica* ISS application shows that pH and carbonate equilibria can dominate interpretation when pH is not directly controlled (Poughon et al. 2021, 8–9).
5. **Harvesting and food-quality gap.** Thin-film productivity must be tied to stable, safe, edible, and processable biomass, not only to dry mass per area.
6. **Reliability gap.** Long-duration life support needs stable sensor calibration, contamination resistance, film morphology control, and recovery from disturbances.

11.2 Recommendations

The strongest near-term study would implement a thin-film algal photobioreactor as a MELiSSA-compatible service module. The model should expose five outputs: oxygen production, carbon dioxide uptake, nitrate uptake, biomass harvest, and risk or fouling state. The model should include a spatial PDE layer for development and a reduced-order control map for operation. It should be benchmarked against MELiSSA standards: mass-balanced equations, identifiable parameters, dynamic disturbance validation, and gas-balance closure. The C3 nitrifying model should be treated as the nitrogen input generator, not as a peripheral subsystem; otherwise a phototrophic panel cannot close the waste-to-food loop.

12 Conclusion

The strongest evidence supports a layered conclusion. The verified astronautical breakthrough is the MELiSSA modeling stack, especially the C4a *Limnospira indica* photobioreactor and the C3 nitrifying fixed-bed biofilm reactor. The strongest strict thin-film mathematics is the terrestrial attached-algal-biofilm and open PDE literature. The best future architecture is not to choose one over the other, but to combine them: use thin-film PDEs to design compact photosynthetic biofilm hardware, then embed reduced-order outputs in MELiSSA-style mass-balanced, dynamically controlled life-support loops.

In practical terms, the first three readings should be: Poughon et al. for the *Limnospira indica* photobioreactor model and ISS/pilot transfer (Poughon et al. 2021, 1–3, 6, 8–9); Pérez et al. and Montràs for the nitrifying fixed-bed biofilm mathematics (Pérez et al. 2005, 2359–2369)(Montràs i Boet 2009, 96–115); and Garcia-Gragera et al. for the integrated pilot plant that turns reactor models into controlled life-support behavior (Garcia-Gragera et al. 2021, 5–6, 11, 13). For strict thin-film algorithmic work, Polizzi et al. and Blanken’s attached-culture work are the most useful next step (Polizzi et al. 2022, 1, 3–5)(Blanken et al. 2014, 2436–2445).

A Implementation sketch for a nitrifying biofilm prototype

The following pseudocode states the local kinetics from equations (24) and (25); a full reactor model then embeds these rates in radial finite volumes and tank-in-series bulk-liquid compartments.

```
def nitrifier_rates(C_NH4, C_NO2, C_O2, X_Nts, X_Ntb,
                  K_NH4=0.0175, K_NO2=5.04e-3,
                  K_O2_Nts=1.616e-4, K_O2_Ntb=5.44e-4,
                  KI_FA_Nts=0.116, KI_FNA_Ntb=1.56e-4,
                  mu_Nts=1.368, mu_Ntb=0.864,
                  I_FA_Ntb=1.0):
    r_Nts = (mu_Nts * X_Nts
             * C_NH4/(K_NH4 + C_NH4 + C_NH4**2/KI_FA_Nts)
             * C_O2/(K_O2_Nts + C_O2))
    r_Ntb = (mu_Ntb * X_Ntb
             * C_NO2/(K_NO2 + C_NO2 + C_NO2**2/KI_FNA_Ntb)
             * C_O2/(K_O2_Ntb + C_O2) * I_FA_Ntb)
    return r_Nts, r_Ntb
```

A method-of-lines implementation would discretize equation (19) into M radial shells for each support bead, compute boundary fluxes to the bulk liquid using equation (26), update biofilm thickness with equation (22), and solve the stiff ODE system with an implicit integrator.

B Reduced-order outputs for a thin-film algal BLSS module

A mission controller does not need the full PDE state at every time step. It needs a compact service vector:

$$\mathbf{y}_{\text{service}} = \begin{bmatrix} \dot{n}_{\text{O}_2} & -\dot{n}_{\text{CO}_2} & -\dot{n}_{\text{NO}_3} & \dot{m}_{X,\text{harvest}} & \rho_{\text{risk}} \end{bmatrix}^T. \quad (49)$$

A reduced-order model can be identified from simulations and experiments:

$$\mathbf{y}_{\text{service}} = \mathbf{W}\Phi(q_0, F_g, Q, h, t_{\text{harvest}}, pH, T) + \boldsymbol{\epsilon}, \quad (50)$$

where Φ is a basis of nonlinear features and ε is model error. The operational rule is then to control the film only inside the validated region:

$$(q_0, F_g, Q, h, t_{\text{harvest}}, pH, T) \in \mathcal{D}_{\text{validated}}. \quad (51)$$

This condition is the thin-film analogue of the MELiSSA photobioreactor feasibility maps and C3 biofilm disturbance-validation logic.

C Citation note

Chicago author-date in-text citations are used throughout. Page locators refer to PDF or article pagination when available. For web-first articles or online materials where conventional pagination is absent, section, abstract, or range locators are used instead of page numbers.

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